Project Overview

Causal Modelling of Player Readiness to Train

KU Leuven & OH Leuven Partnership

---

1. Core Objective

This project develops a prescriptive analytics system that estimates individual player **Readiness to Train** in professional football. Using **Causal Machine Learning** within a **Dynamic Treatment Regime (DTR)** framework, the system recommends an optimal daily training intensity for each player, delivered as a **continuous score between 0 and 1**. A score of 0 indicates the player should rest entirely, while a score of 1 indicates the player can train at maximal intensity. This score simultaneously serves two roles: it is the **recommended treatment** (i.e., the prescribed training intensity from the causal model) and an **observable variable** provided to coaching and medical staff as a decision-support output.

Crucially, the objective is **not** injury minimisation. A policy that minimises injury risk would trivially prescribe zero load, eliminating performance along with risk. The true objective is **performance optimisation under biological constraints**, where some non-zero injury incidence may be optimal because competitive performance requires physiological stress. Injuries are part of the performance-exposure frontier, not a quantity to be driven to zero.

The system is fully **individualised**: each player receives a personalised score derived from their own physiological profile, training history, and evolving state. Because training sessions occur sequentially before each match and each player's history differs, the problem is inherently a **Dynamic Treatment Regime with variable-length decision horizons**.

2. Why This Is a Causal Problem

Standard predictive modelling asks *"given the data, what will happen?"*. This project asks a fundamentally different question: *"given a player's full history up to today, what level of training intensity today best contributes to future match-day performance?"*. Answering this question requires causal reasoning for three interrelated reasons.

2.1 Time-Varying Confounding Affected by Prior Treatment

Training today affects fatigue tomorrow. Fatigue influences what training the coaching staff will prescribe tomorrow. This creates a feedback loop: covariates at time *t* (fatigue, readiness, ACWR) are simultaneously **consequences** of treatment at *t-1* and **causes** of treatment at *t*. Classical regression adjusting for these time-varying confounders would block part of the causal pathway, biasing estimated treatment effects (Robins, Hernan & Brumback, 2000). The system is dynamically endogenous, and standard methods such as propensity-score matching or conventional regression fail because they cannot simultaneously adjust for and not adjust for the same time-varying confounder. This is precisely the setting for which G-methods were developed.

2.2 Treatment-Confounder Feedback

Coaching decisions are observational, not randomised. Hard training sessions are more likely when players appear fresh; recovery sessions are prescribed when fatigue is high; injured players receive systematically different exposures. The observed relationship between training intensity and performance therefore conflates the physiological response to load with the coach's selection behaviour. Disentangling these requires methods that handle this selection-induced covariate shift, such as inverse probability weighting with stabilised weights for continuous treatments (Robins et al., 2000) or parametric G-computation (Hernan & Robins, 2020).

2.3 Sequential Decision-Making

The treatment is not a single static exposure but a repeated, time-varying intervention. A training week typically involves five or six sessions before a match, with each session representing a decision point. The question is not just *"what training today?"* but *"what sequence of training intensities over the cycle maximises match-day performance?"*. Training today shifts the distribution of fatigue, adaptation, and risk tomorrow, which influences both future training decisions and eventual match performance. This is inherently a **Dynamic Treatment Regime** problem (Murphy, 2003; Chakraborty & Moodie, 2013), where the optimal regime is a sequence of decision rules mapping from player state to treatment at each decision point.

3. Causal Structure: Individualised DAGs and Match Cycles

3.1 The Causal Cycle

The causal structure of this problem is best represented through **individualised Directed Acyclic Graphs (DAGs)**. Each player has their own DAG that encodes the relationships between their covariates, treatments, and outcomes. The fundamental unit of analysis is the **match cycle**: the sequence of days between two consecutive matches.

Within each cycle, three groups of variables interact dynamically:

- **Player State (Covariates L_t)**: Subjective wellbeing (fatigue, soreness, sleep quality, stress, mood as individualised Z-scores), workload history (ACWR variables), medical availability, club attendance, composite scores (Physical State, Mental State, Overall Wellbeing), and temporal context (Days Since Game, Days Until Match).
- **Treatment (A_t)**: Daily training intensity, operationalised as a continuous score in [0, 1] derived from GPS-derived workload metrics of that training session. This score aggregates total distance, high-speed running, decelerations, and sprints relative to individual match benchmarks into a single normalised intensity measure.

• **Outcome (Y)**: Match-day performance, measured as an already-constructed variable representing the player's physical intensity per minute played. This is a clear, continuous outcome variable that is available in the dataset and directly reflects in-match performance output.

3.2 Cycle Dynamics and Feedback

The DAG for each player exhibits the following temporal structure within and across cycles:

• Within a cycle, the player state L_t influences treatment assignment A_t (coaches prescribe based on observed state), and treatment A_t causally affects the player state at *t+1* (training changes fatigue, soreness, adaptation).

• At the end of the cycle, the cumulative effect of the treatment sequence manifests in the match-day outcome Y.

• Critically, the match-day outcome Y feeds back into the player state for the **next** cycle. A demanding match increases subsequent fatigue and soreness, while a poor performance may alter coaching strategy. The post-match player state becomes the **baseline state** for the new cycle.

• Across the full season, each player's trajectory can be viewed as **one long causal chain** where L_t influences L_{t+1}, A_t influences L_{t+1}, and match outcomes influence the baseline state of the subsequent cycle. However, the problem is modelled **per cycle** to maintain tractability and align with the natural decision structure of professional football.

3.3 Variable-Length Cycles

A distinctive feature of this setting is that **cycle length varies**. The number of training days between consecutive matches differs due to scheduling (e.g., midweek fixtures, international breaks, cup matches). This means the number of decision points K per cycle is not fixed. The DTR framework must accommodate variable-length treatment sequences, which distinguishes this problem from standard K-stage DTR formulations with fixed K. Methods must either handle variable-length sequences directly or normalise the cycle structure appropriately.

3.4 Within-Row Temporal Semantics

Each row in the dataset represents one player-day at date *t*. Variables within a row have distinct temporal positions, which is critical for avoiding data leakage and correctly specifying causal models:

• **Pre-assessment variables** (fully known at the start of day *t*): Yesterday's activity type, GPS metrics, ACWR values, RPE, and medical/attendance history.

• **Morning assessment** (measured before any activity decision): Wellness Z-scores (fatigue, soreness, sleep quality, stress, mood), medical status, composite state scores, Days Since Game, Days Until Match.

• **Post-assessment** (determined after morning assessment): Activity Type Today (the treatment decision), which is derived from the next row's Activity Type Yesterday. This variable is not available at prediction time and should only be used as a treatment variable.

This temporal ordering is essential: the treatment decision A_t is made *after* observing the player state L_t, and the causal model must respect this ordering.

4. Sport Science Foundations

The dataset design and feature engineering are grounded in established sport science principles that inform both the causal structure and the interpretation of treatment effects.

4.1 Supercompensation

Training stress triggers a temporary performance decline (fatigue) followed by a rebound above baseline, provided adequate recovery. The ACWR (Acute:Chronic Workload Ratio) features are designed to capture where a player sits on the stress-recovery-adaptation curve. This biological mechanism is the fundamental reason why optimal training intensity is neither zero nor maximal: the system must find the intensity that maximises the supercompensation response.

4.2 The Gabbett U-Shaped Risk Model

Following Gabbett (2016), there exists an optimal zone of acute loading (typically ACWR between 0.8 and 1.3) that maximises fitness gains without spiking injury risk. Both chronic underload and acute load spikes are associated with elevated injury probability. This U-shaped relationship between load and injury risk provides sport-science justification for the continuous treatment formulation: the model must learn the player-specific sweet spot along this curve rather than simply categorising load into discrete buckets.

4.3 Individual Profiling

Players have unique physiological baselines, fatigue profiles, and recovery dynamics. Identical absolute loads produce different responses across athletes. The dataset accounts for this through individualised Z-scores (player-specific 28-day rolling windows for subjective wellbeing) and personal match benchmarks (GPS metrics expressed as percentages of each player's own match reference values). This pre-processing embeds within-player normalisation into the feature space, supporting the individualised nature of the DTR.

4.4 Sequential Training Effects

A single training session does not determine match readiness; it is the *sequence* of sessions across the cycle that matters. A hard session followed by adequate recovery produces different outcomes than two hard sessions in succession. This sequential dependence is precisely what the DTR framework captures: the optimal decision at time t depends on the full history of treatments and states up to t.

5. Data

The dataset is a longitudinal, observational panel of **27 professional football players** at OH Leuven, covering **156 days** (approximately one competitive season segment from June to November 2025) and **4,239 player-day observations**. Each row represents one player on one day, combining objective GPS-derived workload metrics, subjective wellbeing indicators, and contextual medical/availability information.

5.1 Dataset Summary

Metric	Value	Detail
Observations	4,239	Player-day records
Variables (raw)	24	GPS, wellbeing, contextual, medical
Variables (processed)	36	After feature engineering
Temporal Span	156 days	Jun-Nov 2025, one season segment
Unique Players	27	Professional squad, OH Leuven
Granularity	Daily	One row per player per day

5.2 Feature Categories

Category	Features	Encoding
External Load (GPS)	Total Distance, High-Speed Distance (>19.8 km/h), Decelerations (>3 m/s2), Sprints	ACWR (7:42 EMA), % of personal match benchmarks
Subjective Wellbeing	Fatigue, Soreness, Sleep Quality, Stress, Mood	Individualised 28-day rolling-window Z-scores
Composite Scores	Physical State, Mental State, Overall Wellbeing	Aggregated from subjective sub-scales
Contextual / Medical	Medical availability %, Club attendance %, Activity reason, Medical status	Categorical (Available / Attention / Injured / Sick / Absent)
Temporal Context	Days Since Game, Days Until Match	Integer; Days Since Game >= 1, Days Until Match >= 0

5.3 Domain-Informed Feature Engineering

The feature space embeds substantial sport science domain knowledge. ACWR variables are computed as 7-day acute over 42-day chronic exponential moving averages, encoding theory about the balance between acute stimulus and chronic adaptation. Performance metrics are expressed as percentages of individual match benchmarks, meaning the data are partially individualised by design. Subjective wellbeing variables are standardised as player-specific Z-scores over 28-day rolling windows, embedding within-player normalisation. This pre-processing means the raw feature space is already richer than typical tabular datasets, but also that domain assumptions are baked into the inputs.

6. Treatment, Outcome, and Observation Structure

6.1 Treatment: Continuous Training Intensity

The treatment is daily training intensity, operationalised as a **continuous variable in [0, 1]**. This score is constructed from the GPS-derived workload metrics of each training session (total distance, high-speed distance, decelerations, sprints), normalised against each player's individual match benchmarks. A score of 0 represents complete rest and a score of 1 represents maximal training intensity equivalent to match-level output.

The continuous treatment formulation is essential for two reasons. First, training intensity in practice is not naturally categorical; it varies along a continuous spectrum and discretising it into categories (e.g., Recovery/Low/Moderate/Hard) discards information and introduces arbitrary boundaries. Second, the continuous formulation aligns with the DTR literature on continuous treatments, enabling the use of methods such as G-estimation with continuous treatments (Robins, 2004) and dynamic weighted ordinary least squares (dWOLS) for continuous treatments (Wallace & Moodie, 2015), which provide doubly robust estimation.

The treatment score also serves as the system's output: the recommended intensity that coaching staff can use to calibrate each player's daily session.

6.2 Outcome: Match-Day Performance Intensity

The outcome is a clearly defined, already-constructed variable in the dataset: **match-day physical intensity per minute played**. This continuous variable captures a player's in-match physical output rate, integrating GPS-derived performance metrics normalised by playing time. It provides a direct, interpretable measure of match-day performance that is available for every player who is selected and participates.

Larger values of the outcome indicate higher match-day performance, which is desirable. The DTR objective is therefore to find the treatment regime (sequence of daily training intensities) that maximises the expected value of this outcome.

6.3 Structural Missingness and Informative Censoring

Several variables exhibit substantial missingness, particularly on days where players did not train, did not play, or were injured. This missingness is **not random** but structurally tied to injury status, selection decisions, and scheduling. Specifically:

- Match-day outcomes are only observed for players who are selected and actually play. Non-selection and injury create **informative censoring** correlated with treatment history.
- GPS training metrics are absent on rest days and for injured players, meaning the treatment variable itself is structurally missing for certain player-days.
- Medical status transitions (Available to Attention to Injured) alter the observed trajectory and must be accounted for methodologically rather than treated as nuisance.

Methods must handle this informative censoring explicitly. G-computation naturally accommodates this by modelling the full joint distribution of post-treatment variables, while inverse probability of censoring weights (IPCW) can be combined with treatment weights in the MSM framework (Hernan, Brumback & Robins, 2001).

7. Statistical Regime and Methodological Challenges

The dataset occupies a distinctive **"Small N, Large T"** statistical regime: only 27 players but moderately long time series per player. This has direct consequences for modelling strategy and defines several key challenges.

7.1 Data Efficiency over Model Complexity

High-capacity, data-hungry models cannot rely on massive between-player heterogeneity. The signal is likely dominated by within-player dynamics, and models must either be parsimonious or leverage **partial pooling** across individuals to avoid overfitting. The DTR estimation methods must balance individual-level tailoring with the limited sample size.

7.2 Observational Confounding

Because treatment assignment is non-randomised and driven by the coaching staff's implicit policy, all observed training-performance associations are confounded by selection behaviour. Methods must explicitly handle this covariate shift. For continuous treatments, the generalised propensity score (GPS) replaces the binary propensity score, and stabilised continuous weights are used to create a pseudo-population in which treatment is independent of measured confounders (Robins et al., 2000; Hernan & Robins, 2020).

7.3 Sequential Treatment Effects with Variable-Length Horizons

The causal question concerns the effect of treatment *sequences*, not single exposures. Moreover, cycle lengths vary between matches, creating variable-length decision horizons. This requires methods from the dynamic treatment regime literature that can handle K-stage problems where K is not fixed across cycles. Standard cross-sectional treatment effect estimators (including meta-learners such as the S-Learner, T-Learner, or X-Learner) are **not appropriate** for this setting because they do not account for time-varying confounding affected by prior treatment, nor do they optimise over sequential decisions.

7.4 Informative Censoring and Selective Observation

Match-day outcomes are only observed for selected, available players. Injury and non-selection create censoring processes correlated with treatment history, introducing survivorship bias if ignored. The analysis must either model the censoring mechanism explicitly or weight observations to account for selective observation.

7.5 Causal Identification Assumptions

Identification of causal effects in this setting relies on three core assumptions from the potential outcomes framework, which must be carefully evaluated:

- **Sequential exchangeability (no unmeasured confounding)**: At each time point, treatment assignment is independent of potential outcomes conditional on the observed history. This is plausible given the rich set of measured covariates (GPS metrics, wellbeing scores, medical status), but unmeasured factors (e.g., personal stressors, tactical considerations) could threaten this assumption. Sensitivity analyses should assess robustness.
- **Positivity (overlap)**: For every possible covariate history, there must be a positive probability of receiving any treatment level. With continuous treatment, this requires the conditional density of

treatment given history to be positive over the relevant support. Practically, this means we need sufficient variation in prescribed training intensities across different player states.

• **Consistency**: The observed outcome under the treatment actually received equals the potential outcome under that treatment. For continuous treatments, this requires a well-defined treatment variable without hidden treatment-variation irrelevance issues.

8. Methodological Approach: Dynamic Treatment Regimes

The project adopts a **Dynamic Treatment Regime (DTR)** framework as the primary methodological approach. A DTR is a sequence of decision rules d = (d_1, ..., d_K), one per decision point, that maps from a player's evolving covariate and treatment history to an optimal treatment action (Murphy, 2003; Chakraborty & Moodie, 2013). The goal is to estimate the **optimal DTR**: the regime that, if followed for the entire population, would maximise expected match-day performance.

8.1 G-Methods for Time-Varying Confounding

The backbone of the causal estimation strategy is the family of **G-methods** (Robins, 1986), specifically designed for settings with time-varying confounders affected by prior treatment. Three principal G-methods are considered:

G-Computation (Parametric G-Formula)

G-computation models the joint distribution of all post-treatment variables (covariates, treatments, and outcomes) through a sequence of conditional models. It then simulates counterfactual outcomes under hypothetical treatment regimes by intervening on the treatment variable at each time point while propagating the effects through the modelled system. This approach naturally handles the feedback loop between treatment and time-varying confounders because it separately models each component of the joint density, avoiding the collider-stratification bias that plagues standard regression. For continuous treatments, the G-formula can integrate over the treatment density to estimate the expected outcome under any specified regime.

Marginal Structural Models with Inverse Probability of Treatment Weighting

Marginal Structural Models (MSMs; Robins, Hernan & Brumback, 2000) specify a parametric model for the marginal mean of the potential outcome as a function of treatment history. Parameters are estimated using Inverse Probability of Treatment Weighting (IPTW), which creates a pseudo-population in which treatment is independent of measured confounders. For **continuous treatments**, the weights are ratios of marginal to conditional treatment densities: $sw_t = f(A_t) / f(A_t \mid \text{L-bar}_t, \text{A-bar}_{t-1})$, where stabilised weights are used to reduce variance. MSMs are particularly attractive here because they provide a single parsimonious model for the causal effect of the full treatment sequence on the match-day outcome.

G-Estimation for Structural Nested Models

G-estimation targets the parameters of a Structural Nested Mean Model (SNMM), which directly models the **blip function**: the additional effect of the treatment actually received at time *t* versus a reference treatment, conditional on history. G-estimation is attractive because it is robust to misspecification of the treatment model (as long as the blip model is correctly specified) and naturally accommodates continuous treatments. It is closely related to the optimal DTR estimation because the optimal decision at each stage is the treatment that maximises the blip function.

8.2 DTR Estimation Methods

Building on the G-methods for unbiased effect estimation, the project considers the following approaches for estimating the **optimal treatment regime**:

Q-Learning

Q-learning (Watkins, 1989; Murphy, 2005) is a regression-based backward induction method. Starting from the final stage K, it models the expected outcome as a function of state and treatment (the Q-function), then works backward to earlier stages, at each stage substituting the future value under the estimated optimal policy. The optimal treatment at each stage is the value that maximises the Q-function. For continuous treatments, this involves optimising over the treatment space rather than simply comparing discrete options.

Dynamic Weighted Ordinary Least Squares (dWOLS)

dWOLS (Wallace & Moodie, 2015) combines the simplicity of Q-learning with the doubly robust property of G-estimation. It estimates optimal DTRs through a sequence of weighted regressions, where the weights are functions of the generalised propensity score. dWOLS is **doubly robust**: consistent estimates are obtained if either the treatment model or the outcome model is correctly specified, but not necessarily both. Crucially, dWOLS has been extended to **continuous treatments** (Wallace & Moodie, 2015) and to **time-to-event outcomes** (Simoneau et al., 2020), making it well-suited for this application.

Off-Policy Reinforcement Learning

As a complementary direction, off-policy reinforcement learning (RL) methods can learn optimal sequential policies directly from observational data. Fitted Q-Iteration and batch policy gradient methods can optimise the full sequence of treatment decisions without requiring backward induction. However, RL methods typically demand larger sample sizes than regression-based DTR methods, and the small N in this setting (27 players) may limit their applicability. They remain a secondary methodological avenue, potentially viable if partial pooling across players provides sufficient effective sample size.

8.3 Method Selection Rationale

The following table summarises the candidate methods and their suitability for this specific problem:

Method	Handles TVConf	Continuous Tx	Sequential Opt.	Small N Feasibility
G-Computation	Yes	Yes	Via simulation	Good (parametric)
MSM / IPTW	Yes	Yes (density weights)	Via MSM specification	Good (parametric)
G-Estimation / SNMM	Yes	Yes	Via blip maximisation	Good (semi-parametric)
Q-Learning	Yes (backward induction)	Yes	Yes (native)	Moderate
dWOLS	Yes (doubly robust)	Yes (extended)	Yes (native)	Good
Off-Policy RL	Yes	Yes	Yes (native)	Limited (data-hungry)

TVConf = time-varying confounding affected by prior treatment; Continuous Tx = continuous treatment support; Sequential Opt. = native support for sequential optimisation.

Note that **causal meta-learners** (S-Learner, T-Learner, X-Learner, DR-Learner) are **not suitable** for this problem. These methods are designed for single-stage treatment effect estimation and do not handle time-varying confounding affected by prior treatment. They estimate conditional average treatment effects (CATEs) at a single point in time, not optimal treatment sequences. Similarly, standard survival analysis models, while useful for handling censoring, do not directly address the sequential optimisation problem.

9. Practical Application

The end product is a **daily decision-support tool** for coaching and medical staff. Each morning, after the player's wellness assessment is recorded, the system produces a personalised **continuous readiness score between 0 and 1** for each player. This score represents the optimal training intensity for that day, accounting for the player's full history within the current match cycle and optimising toward match-day performance.

9.1 Interpreting the Score

Score Range	Interpretation	Practical Guidance
0.0 - 0.2	Rest or minimal activity	Player is approaching or in a high-risk zone. Significant load reduction or complete rest advised.
0.2 - 0.4	Light recovery session	Active recovery recommended. Low-intensity movement to promote adaptation without adding stress.
0.4 - 0.6	Moderate training	Balanced training load. Player can participate in tactical and technical work at moderate intensity.
0.6 - 0.8	High-intensity training	Player's adaptation trajectory is positive. Can sustain demanding training.
0.8 - 1.0	Maximal training intensity	Player is in optimal condition. Full participation in the most demanding sessions.

The continuous score preserves the full granularity of the model's recommendation, avoiding the information loss inherent in categorical systems. Coaching staff can use it directly to calibrate session design, player groupings, and individual load targets. The score also serves as the treatment variable in the dataset, creating a closed-loop system where the model's output feeds directly into the causal framework.

9.2 Individualisation

Every aspect of the system is individualised. Each player's DAG, feature encoding (via player-specific Z-scores and personal benchmarks), treatment response model, and optimal policy are tailored to their unique physiological profile. The DTR framework naturally supports this: the decision rule at each stage is a function of the individual player's own covariate and treatment history, not a population average.

10. Summary

This project addresses a longitudinal, observational, dynamically confounded panel of elite athletes, where daily training intensity is repeatedly assigned based on evolving player state. The central object of interest is an **individualised, optimal Dynamic Treatment Regime**: a sequence of decision rules mapping each player's current and historical state to an optimal continuous training intensity score in [0, 1].

The system exhibits: (1) time-varying confounding affected by prior treatment, (2) endogenous treatment assignment via coaching decisions, (3) sequential treatment effects with variable-length cycles, (4) informative censoring of match-day outcomes, (5) feedback from match outcomes to subsequent player state, and (6) strong domain-informed feature engineering grounded in sport science.

The methodological approach centres on **G-methods** (G-computation, MSMs with IPTW, G-estimation) for unbiased estimation under time-varying confounding, combined with **DTR estimation methods** (Q-learning, dWOLS) for sequential optimisation of continuous treatments. The output is a personalised daily readiness score that serves as both the recommended training intensity and a causal treatment variable, delivered to coaching staff through a practical decision-support interface.

Solving this problem requires methods at the intersection of causal inference, sequential decision-making, and sports science domain knowledge.

References

- Chakraborty, B. & Moodie, E. E. M. (2013). *Statistical Methods for Dynamic Treatment Regimes.* Springer.
- Gabbett, T. J. (2016). The training-injury prevention paradox. *British Journal of Sports Medicine, 50*(5), 273-280.
- Hernan, M. A. & Robins, J. M. (2020). *Causal Inference: What If.* Chapman & Hall/CRC.
- Hernan, M. A., Brumback, B., & Robins, J. M. (2001). Marginal structural models to estimate the joint causal effect of nonrandomized treatments. *Journal of the American Statistical Association, 96*(454), 440-448.
- Murphy, S. A. (2003). Optimal dynamic treatment regimes. *Journal of the Royal Statistical Society: Series B, 65*(2), 331-355.
- Murphy, S. A. (2005). A generalization error for Q-learning. *Journal of Machine Learning Research, 6*, 1073-1097.
- Robins, J. M. (1986). A new approach to causal inference in mortality studies with sustained exposure periods. *Mathematical Modelling, 7*(9-12), 1393-1512.
- Robins, J. M. (2004). Optimal structural nested models for optimal sequential decisions. In D. Y. Lin & P. Heagerty (Eds.), *Proceedings of the Second Seattle Symposium on Biostatistics* (pp. 189-326). Springer.
- Robins, J. M., Hernan, M. A., & Brumback, B. (2000). Marginal structural models and causal inference in epidemiology. *Epidemiology, 11*(5), 550-560.
- Schulte, P. J., Tsiatis, A. A., Laber, E. B., & Davidian, M. (2014). Q- and A-learning methods for estimating optimal dynamic treatment regimes. *Statistical Science, 29*(4), 640-661.
- Simoneau, G., Moodie, E. E. M., Nijjar, J. S., & Platt, R. W. (2020). Estimating optimal dynamic treatment regimes with survival outcomes. *Journal of the American Statistical Association, 115*(531), 1531-1539.
- Wallace, M. P. & Moodie, E. E. M. (2015). Doubly-robust dynamic treatment regimen estimation via weighted least squares. *Biometrics, 71*(3), 636-644.
- Wang, C. et al. (2025). A tutorial on optimal dynamic treatment regimes. *arXiv:2502.16988*.
- Watkins, C. J. C. H. (1989). *Learning from Delayed Rewards.* PhD thesis, King's College, Cambridge.